KAI SANDAI HACKATHON · AGRITECH TRACK

KrishiMitra AI

The AI friend in every farmer's pocket: diagnose crops, beat the weather and get fair prices, in their own language.

Diagnose · Predict · Price · Speak

Presented by [Your Team Name]

AGENDA

From Seed to Harvest: Our Pitch Today



PART 1 · THE SOIL

The Problem

01 The Problem

02 By the Numbers

03 Meet Ramu



PART 2 · THE SEED

Our Solution

04 Our Solution

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PART 3 · THE HARVEST

The Impact

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Farmers Feed 1.4 Billion People, but Fight Alone

“A farmer makes hundreds of decisions every season, with almost no expert to ask.”

Small farmers lose money at every step, from sowing to selling. One wrong guess can cost a whole season's income.

Crop Disease & Pests

Diseases are spotted too late, and pesticides are bought on guesswork.

Unpredictable Weather

Unseasonal rain and heatwaves wipe out months of work overnight.

Middlemen & Low Prices

Farmers rarely know live mandi rates, so they sell cheap.

Language & Access Gap

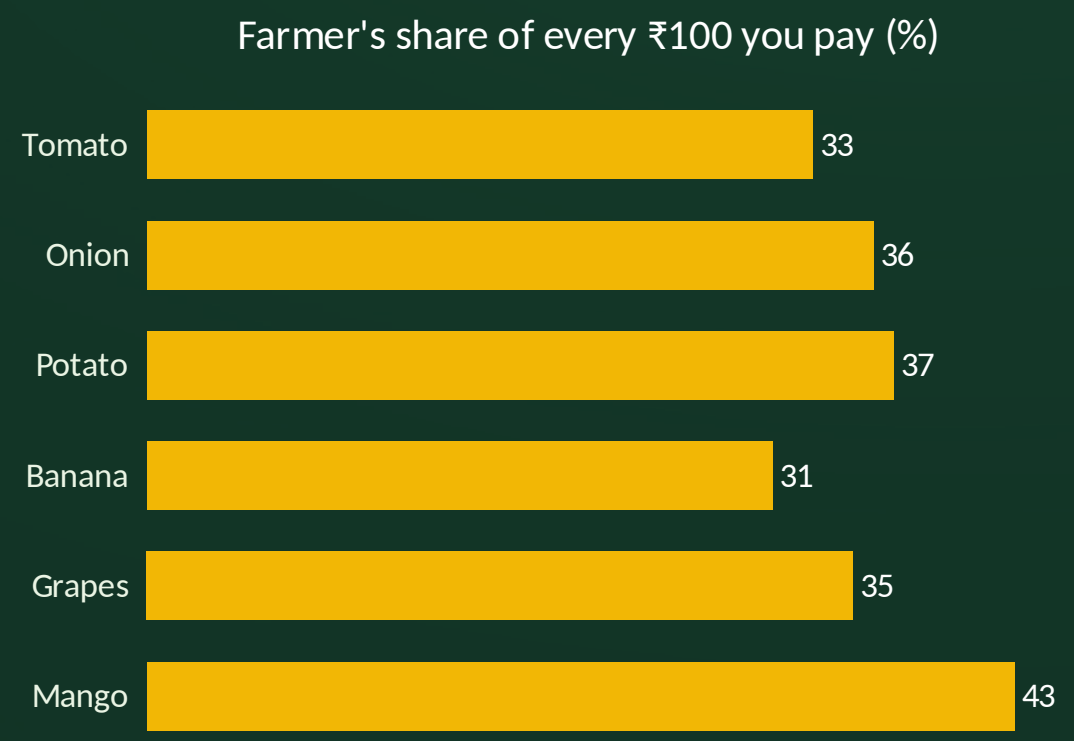
Expert advice is in English, online or 40 km away.

The Scale of the Problem

86% of India's farmers are small and marginal, farming less than 2 hectares

46% of India's workforce depends on agriculture for a living

40% of global crop output can be lost to pests and diseases each year



Sources: Agriculture Census 2015-16 · PLFS 2023-24 · FAO · RBI Working Paper (2024)

One Farmer, One Season, Many Losses



Ramu, 42

Cotton & soybean farmer
2 acres · Vidarbha, Maharashtra
Speaks Marathi · basic Android phone

ILLUSTRATIVE PERSONA

One season in Ramu's life

WEEK 6
White spots appear on his leaves. The nearest agri expert is 40 km away.

WEEK 7
The local shopkeeper sells him the wrong and costlier pesticide.

WEEK 14
Unexpected rain ruins the harvest he left drying in the open.

HARVEST
He sells to a middleman below the mandi rate. He never saw the live prices.

The result: higher costs, lower yield and a lower price, every single season.

KrishiMitra AI: An Expert in Every Pocket

Snap & Diagnose

Photo of a leaf gives the disease and its cure.

Ask by Voice

Speak in your own language and get answers.

KrishiMitra AI

1 app · 4 superpowers

Weather Shield

Hyperlocal alerts: when to spray and sow.

Fair Price Finder

Live mandi prices and the best place to sell.

Available on Android · WhatsApp · Voice call (IVR) · SMS

Everything a Farmer Needs, in One Place

Crop Doctor

A leaf photo gives the disease, its severity and the right treatment in seconds.

Voice Assistant

Ask anything by speaking in 10+ Indian languages. No typing needed.

Weather Shield

Hyperlocal forecasts turned into clear spray, irrigate or harvest advice.

Mandi Prices

Live prices from nearby markets, plus the best day and place to sell.

Scheme Finder

Matches each farmer to schemes like PM-KISAN and PMFBY crop insurance.

Offline & Low-Data

Works on 2G, SMS and IVR calls, so even basic phones are covered.

From a Photo to a Decision in Seconds

01

Snap or Speak

Farmer sends a leaf photo or a voice question.

02

AI Understands

Multimodal AI reads the image, voice and location.

03

Smart Advice

Combines weather, soil and price data into one answer.

04

Farmer Gains

Acts on time: saves the crop, the cost and the price.

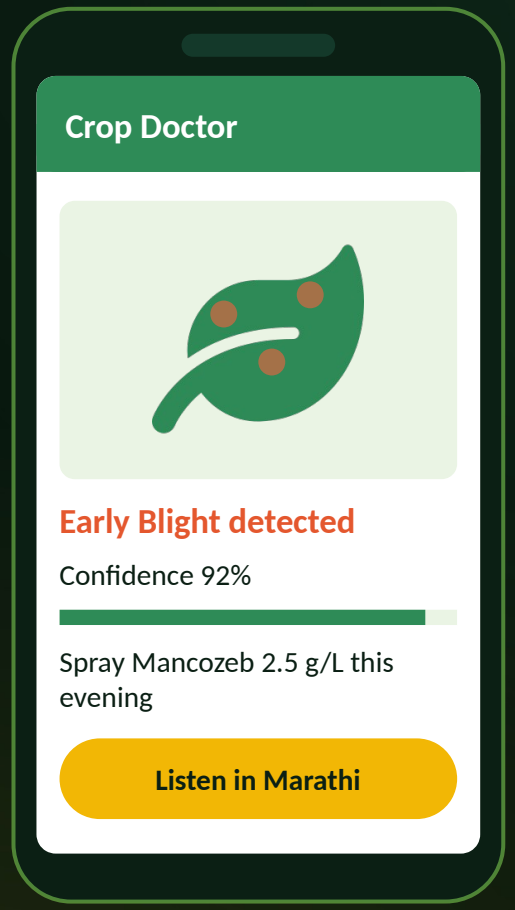
REAL-WORLD EXAMPLE

Ramu sends a cotton leaf photo and asks in Marathi. **KrishiMitra replies: “Bacterial blight detected (92% sure). Spray copper oxychloride 3 g/L this evening; no rain is expected for 48 hours.”**

Built on Open Data and Multimodal AI

Access Layer How farmers reach us	Android App	WhatsApp Bot	IVR Voice Call	SMS Alerts
AI Engine The brain	Gemini Multimodal LLM	Crop Disease Vision Model	Indic Speech-to-Text & TTS	Recommendation Engine
Data Layer Trusted public sources	IMD Weather	Agmarknet Mandi Prices	Soil Health Card	Govt Schemes DB
Cloud & Ops Scale & insights	Cloud Run + Firebase	Vector DB (Agri Knowledge)	FPO Analytics Dashboard	Privacy & Consent Layer

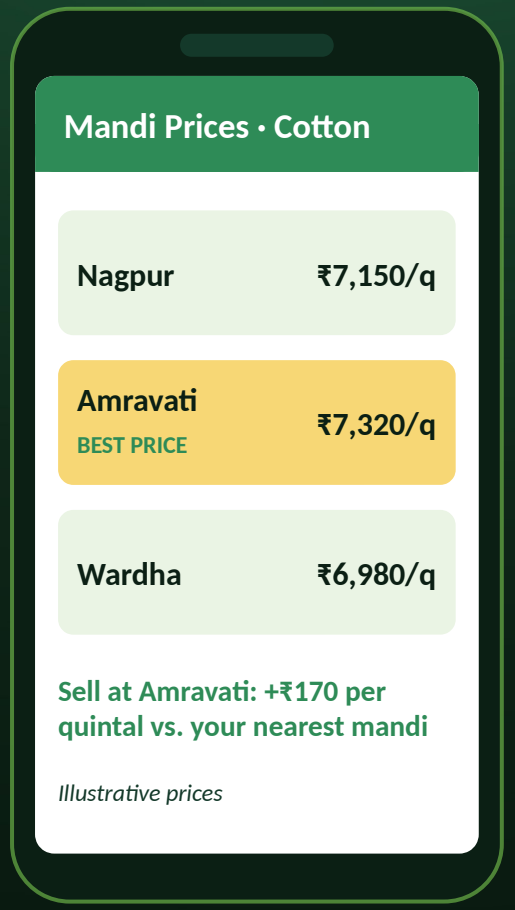
Simple Enough for Every Farmer



1 · Snap a leaf



2 · Ask in your language



3 · Sell at the best price

What Success Looks Like

30%
less crop loss through early disease detection

20%
lower spend on pesticides and fertiliser

15%
better price by selling at the right mandi

1M
farmers reached within 18 months

Pilot targets, to be validated in our 500-farmer field pilot



For the Farmer

More income, less risk, and decisions made in their own language.



For the Planet

The right chemical dose, smarter water use and healthier soil.



For the Nation

Stronger food security that advances SDG 2: Zero Hunger.

What Makes KrishiMitra Different

Capability	Generic Agri Apps	Kisan Call Centre	KrishiMitra AI
Photo-based disease diagnosis	Partial	No	Yes
Voice, in the farmer's own language	Partial	Yes	Yes
Advice tailored to farm, weather & soil	Partial	Partial	Yes
Live mandi prices + best place to sell	Partial	Partial	Yes
Instant answers, 24 × 7	Yes	Partial	Yes
Works without a smartphone	No	Yes	Yes



Our edge: one assistant that sees, listens and speaks the farmer's language.

Free for Farmers, Sustainable by Design



₹0

for farmers. Always.

Core advisory, alerts and prices stay free: trust first, scale second.

FPOs & Co-operatives

A dashboard to advise and track thousands of member farmers.

SUBSCRIPTION

Agri-Input & Insurance Partners

Verified product listings and consent-based, anonymised crop-risk insights.

PARTNERSHIP

Government & CSR

White-label deployments for state agriculture departments.

CONTRACTS

From Seed to Harvest: Our Growth Plan

NOW · SEED

Hackathon MVP

- Crop Doctor for 5 crops
- Voice in 3 languages
- Live mandi prices

MONTH 3 · SPROUT

Field Pilot

- 500 farmers, 2 districts
- Partner with 1 FPO
- Measure yield & income

MONTH 6 · GROW

Expand

- 10 Indian languages
- IVR + offline mode
- Scheme finder live

MONTH 18 · HARVEST

Scale

- 1 million farmers
- 5 states
- Govt & insurer tie-ups

Thank You

From seed to sale, no farmer left alone.

Questions? Let's grow together.



JAI KISAN

[Your Team Name] · KrishiMitra AI · Kai Sandai Hackathon